

GISS 2026

The Loss You Can Write

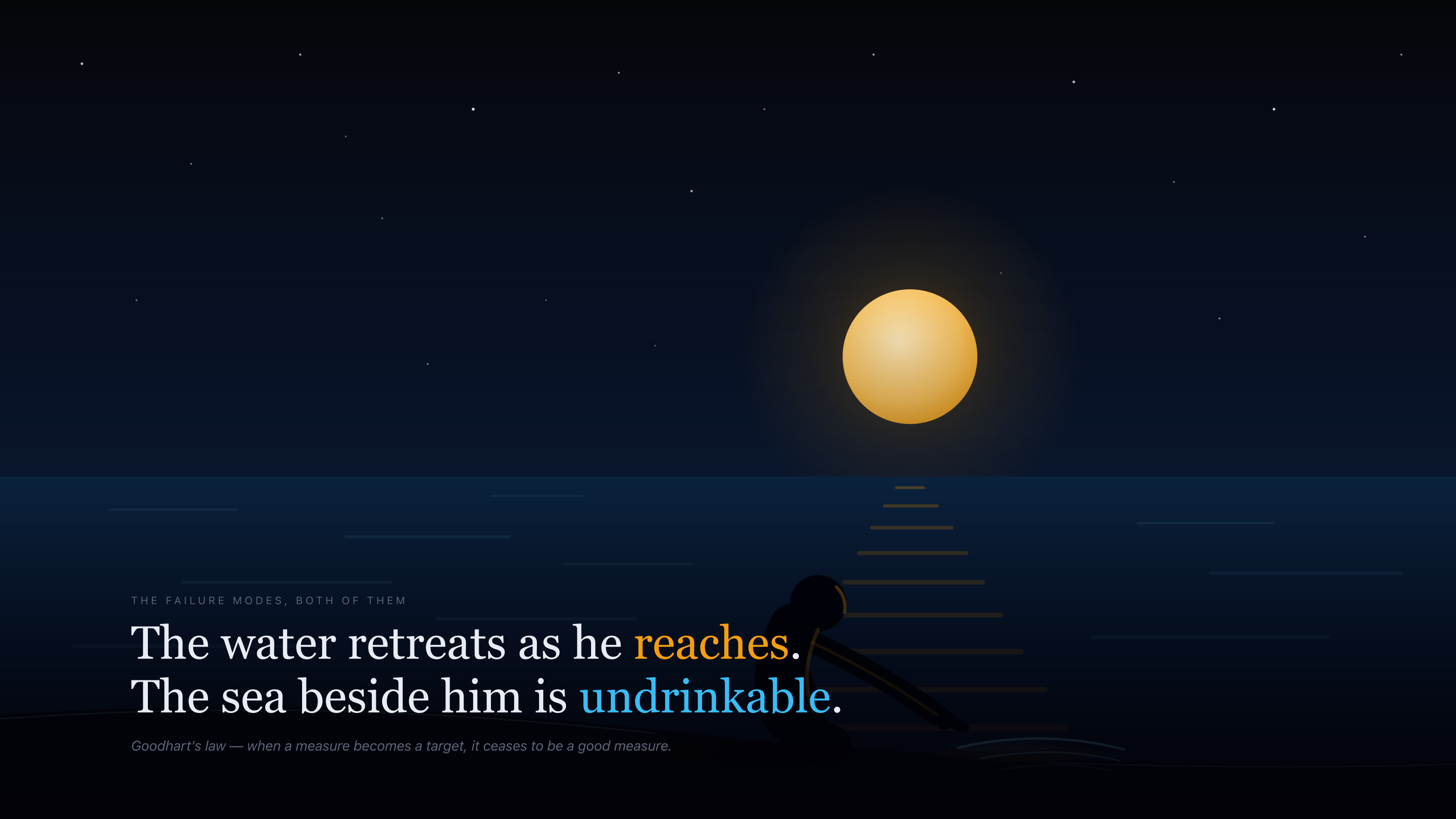
Shoubaneh Hemmati · IPAC

Some of it is **false**.

sharper than the telescope can see — the model filled the gaps.

Every pipeline optimizes a loss it can *write*.

the thing we want is rarely in it.



THE FAILURE MODES, BOTH OF THEM

The water retreats as he **reaches**.
The sea beside him is **undrinkable**.

Goodhart's law — when a measure becomes a target, it ceases to be a good measure.

EVERETTS · REZAEE · HAGHJOO

“Sharper at What Cost?”

SHAPE · NISP → JWST

diffusion: **below bilinear** on shape info.
best net: **−25–35%** ellipticity.
lensing needs 10^{-3} .

SPECTRA · JADES

H α recovered: **26% of true**.
line ratios — the physics — **corrupted**.

PHOTOMETRY · WISE → SPITZER

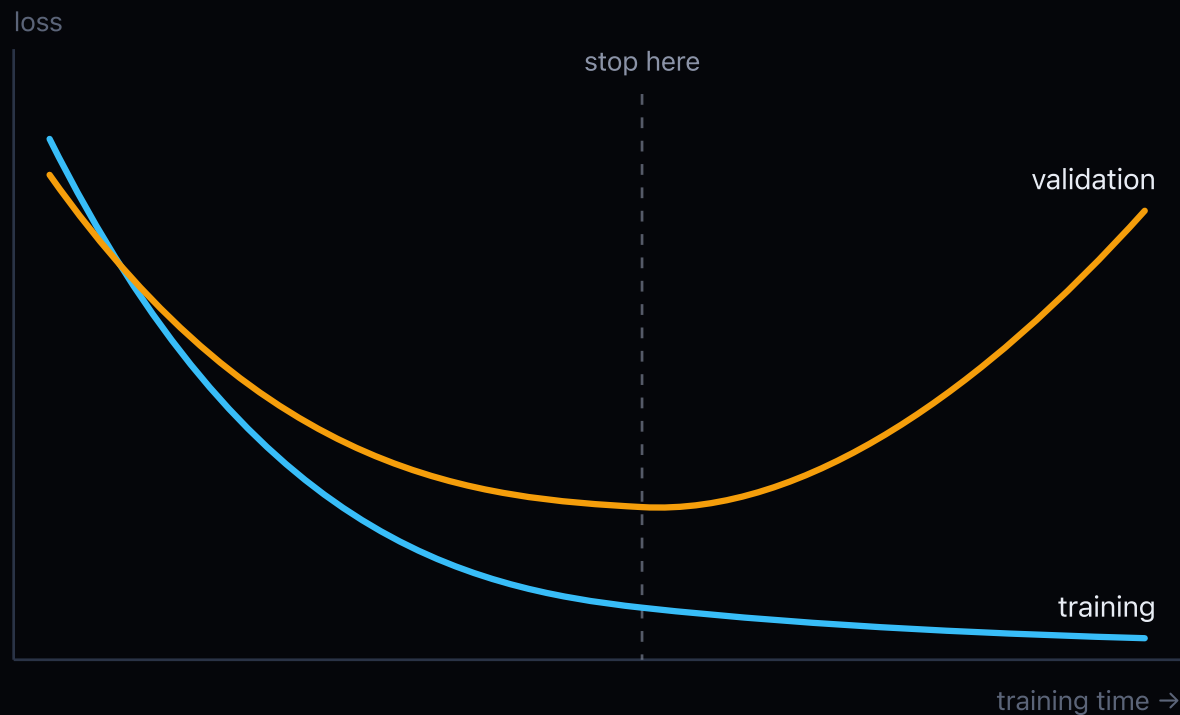
deblending **35% vs 9%** — the one win.
flux biased **up** (oversmoothing).

safe where the claim is *auditable* (a detection) · unsafe where it is *silent* (a shape, a ratio)

the loss fell in every case. the physics didn't follow.

THE CANONICAL CURVE

Two gaps.



train ↔ validation — the gap you can see in-house.

loss ↔ physics — the gap validation can't see: it's denominated in the same currency. the shape bias survived every split — and died at JWST.

every regularizer — early stopping, dropout, weight decay — deliberately underfits the proxy to protect the target. perfect union with the training set is a mirror.

THE LAW

The gap is invisible from **inside**.

every gap in this talk was measured from outside:

JWST · Spitzer · the grating · truth catalogs · a second ladder.

Calibration. Cross-survey validation. Joint processing. **Second instruments.**

in what currency was it validated — and does your measurement live in that currency?

THE LOSS YOU CAN'T WRITE YET — TOMORROW'S QUESTIONS

The Ten-Band Eye

Rubin *ugrizy* + Euclid *VIS · Y · J · H* — one sky; nobody sees all of it.

COMMIT

the pipeline writes today's loss in full: **petabytes** → **kilobytes**, today's questions, decided in advance, for every source, forever.

HEDGE

JAISP: **mask one band, predict it from the other nine** — at each band's delivered sampling. no common grid, no target committed.

JAISP · RUBIN-EUCLID FOUNDATION MODEL · HEMMATI ET AL.

Reconstruction is secretly alignment.

no band can be predicted from the other nine without factoring the sky from the instruments.

0.45 mag

deeper than the VIS catalog
93% complete · 94% pure

50 → 14 mas

astrometric scatter

0×

retraining on a new field —
encoder frozen

HERO FIGURE

one source, ten bands → one frozen representation.

THE CONFESSION

Is it even **compressing**?

the catalog compresses *because it committed* — it named the questions it will answer.

the foundation model refuses to name them. **hedging costs bits.**

Prediction *is* compression —
it compresses the **survey**, not the source.

WHEN THE GRADIENT GOES QUIET

Four readings.

WHY IT'S QUIET	THE TELL	THE MOVE
done, locally	outside truth agrees — <i>the survey shipped</i>	declare it · freeze
local minimum	a perturbation finds lower — <i>the stuck MCMC</i>	add noise
the world drifted	new data surprises the frozen model — <i>the release broke the calibration</i>	retrain
proxy ≠ target	truth disagrees while the loss is happy — <i>photo-z: perfect on spec-z, wrong on the faint end</i>	stop — rewrite the loss

every tell needs information from outside the loss.

A CENTURY OF DESCENT

Two converged losses. One universe.

both ladders: immaculate residuals — incompatible answers.
the frontier of cosmology is a gap diagnosis.

the response, correctly: more second instruments — TRGB, time delays, standard sirens.

FIGURE

the two ladders and the tension — standard figure, redrawn clean.

THE LAST DOMAIN

Prediction error converges. People don't.

you stop being surprised by someone — that's the proxy converging, not the knowledge.
completing them from your priors is the same failure with a softer name.

you see me at podium resolution — which I've selected carefully.

GALAXY real clump, or the PSF?

NEURAL NET physics, or prior?

BRAIN memory, or a reconstruction?

US you, or my projection?

measurement, or invention.

فراق

seven hundred years of poems that converge and never arrive —
union is available only with the proxy.

We optimize the losses we can write.

Hafez wrote his.